

Major Project · TY Computer Engineering

VisionAid

A deep learning-based assistive system for visually impaired people — real-time object detection, text reading, and audio feedback from a single smartphone camera.

Five ways this system moves past existing apps like Seeing AI and Envision:



Works fully offline

Object detection, text recognition, and speech all run **on the phone itself** — no internet connection, no cloud fees, no lag waiting on a server.



Speaks your language

Reads signs and documents aloud in **Hindi and Marathi**, not just English — built for Indian users first, not adapted as an afterthought.



Warns what matters most

An obstacle in your path is announced **immediately and louder**; a poster on a wall waits its turn — feedback ranked by real urgency, not read in a flat list.



Tells you the direction

Stereo audio cues mean "chair on your left" **actually sounds like it's on your left** — spatial feedback most existing apps skip.



Runs on any phone

Optimized for **budget Android devices**, not just flagships — because affordable hardware is the whole point of accessible AI.

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Major Project — First Review